Oracle Utilities Customer Care and Billing v25.10HOT

Product Fix Design

Bug 39064768

Last Modified: March 13, 2026 7:42 AM
Printed On: November 4, 2009

Oracle Utilities Customer Care and Billing Product Fix Design

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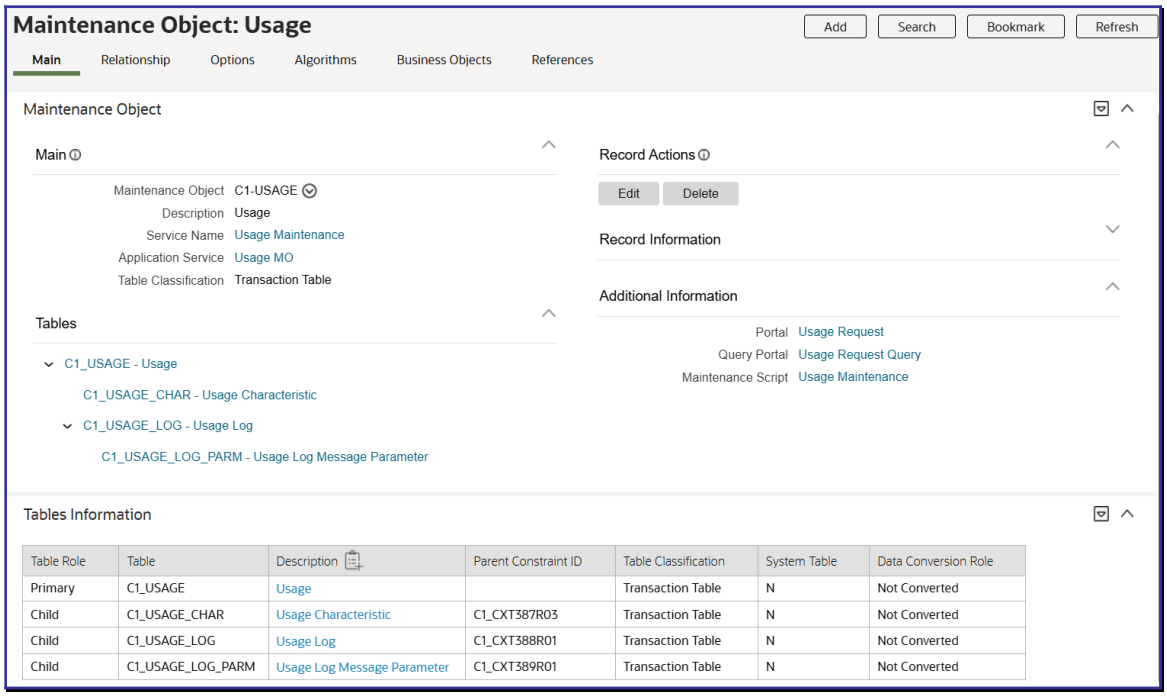
Reported Issue

Description

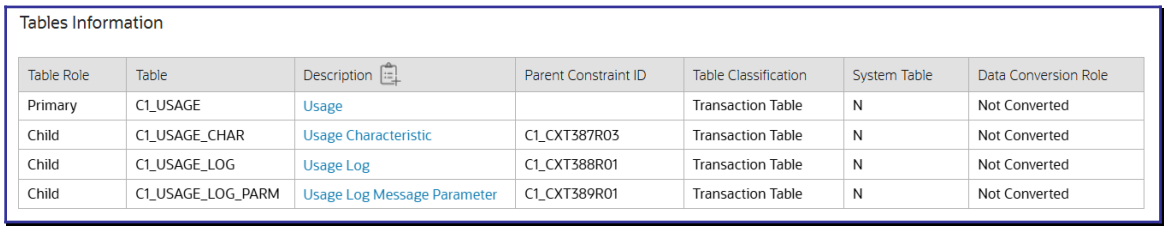
Unable to migrate data for maintenance object **Usage** ***(C1-USAGE)*** due to invalid values for the data conversion roles of its tables.

Steps To Recreate

- Go To Maintenance Object
- Search for **Usage** ***(C1-USAGE)***



Maintenance Object - C1-USAGE



C1-USAGE – Table Information

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Expected Result: The tables that are needed to be converted has Data conversion role of **'Convert'.**

Actual Result: The Data conversion role is **'Not Converted'.**

Solution Design

- Update the Data Conversion Role value of the **Usage** ***(C1_USAGE)*** primary table from 'Not Converted' to 'Convert (New PK)' so that the created Batch Controls can process the table properly.
- Update the Data Conversion Role value of the **Usage Characteristic** ***(C1_USAGE_CHAR)*** child table from 'Not Converted' to 'Convert (Retain PK)' so that the created Batch Controls can process the table properly.
- Update the Data Conversion Role value of the **Usage Log** ***(C1_USAGE_LOG)*** child table from 'Not Converted' to 'Convert (Retain PK)' so that the created Batch Controls can process the table properly.
- Update the Data Conversion Role value of the **Usage Log Message Parameter** ***(C1_USAGE_LOG_PARM)*** child table from 'Not Converted' to 'Convert (Retain PK)' so that the created Batch Controls can process the table properly.
- Create the Conversion Batch Controls by using the Maintenance Object Configuration

Summary

Change Type	Y/N
Does this fix require any Admin (Soft) Table Changes to be made by the client?	N
Does this fix require upgrade data changes to be applied?	N
Does the fix involve DB Changes?	N
Is there an impact on any user interface or any other on-line behavior?	N
Is there an impact on any batch process?	**Y**
Is there an impact on CM – Custom Modification processing?	N
Does this fix require an application viewer update for Algorithm data (run F1-AVALG)?	N
Does this fix require an application viewer update for MO data (run F1-AVMO)?	N
Does this fix require an application viewer update for Table data (run F1-AVTBL)?	N

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Other Changes

Tables

Usage (C1_USAGE) (Existing)

Properties

Property	Value
Description	Usage
Table Type	Table
Table Classification Type	Transaction Table
Data Conversion Role	~~Not Converted~~ Convert (New PK)
Maintenance Object	Usage
Type of Key	System- Generated

Usage Characteristic (C1_USAGE_CHAR) (Existing)

Properties

Property	Value
Description	Usage Characteristic
Table Type	Table
Table Classification Type	Transaction Table
Data Conversion Role	~~Not Converted~~ Convert (Retain PK)
Maintenance Object	Usage
Type of Key	Other

Usage Key (C1_USAGE_K) (Existing)

Properties

Property	Value
Description	Usage Key
Table Type	Table
Table Classification Type	Transaction Table
Data Conversion Role	~~Not Converted~~ Convert (Retain PK)

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Usage Log (C1_USAGE_LOG) (Existing)

Properties

Property	Value
Description	Usage Log
Table Type	Table
Table Classification Type	Transaction Table
Data Conversion Role	~~Not Converted~~ Convert (Retain PK)
Maintenance Object	Usage
Type of Key	Other

Usage Log Message Parameter (C1_USAGE_LOG_PARM) (Existing)

Properties

Property	Value
Description	Usage Log Message Parameter
Table Type	Table
Table Calssification	Transaction Table
Data Conversion Role	~~Not Converted~~ Convert (Retain PK)
Maintenance Object	Usage
Type of Key	Other

Batch Controls

Validate C1-USAGE (C1USAGEV) (New)

- After creating Batch Controls using the Maintenance Object Configuration, the **Entity Validation Template** (***F1-CVVAL***) Batch Control will be duplicated for the Usage MO.

Properties

Property	Property Value
Batch Control	C1USAGEV
Description	Validate C1-USAGE
Template	Entity Validation Template
Selection Algorithm	Entity Validation Select Records
Processing Algorithm	Entity Validation Process Record

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Detailed Description	This batch process validates entities for the specified maintenance object. By default all records are selected and each entity is fully validated for schema as well as object validation rules, at the maintenance object and business object level. The "Process Every Nth" parameter may be used to randomly select a subset of entities to validate. Processing may be further restricted to schema validation only or entity validation rules using the "Validation Mode" parameter. These options may be useful in identifying common data issues in preliminary runs. Errors encountered during validation are logged onto the conversion error table. Note that at the start of this job, all rows in the conversion error table for the process maintenance object are deleted. This process is multi-threaded. Records between commits: N/A Minutes between cursor re-initiation: N/A

Parameters

Sequence	Parameter Name	Description	Required	Parameter Value
110	mo	Maintenance Object	Y	C1-USAGE
120	validateMode	Validation Mode	N	
130	everyNth	Process Every Nth	N	
900	emailMode	Email Mode	N	
910	DIST-THD-POOL	Thread Pool Name	N	
920	maxErrors	Override maximum errors	N	

Resolve XML for C1-USAGE (C1USAGER) (New)

- After creating Batch Controls using the Maintenance Object Configuration, the **XML Resolution Template** (***F1-CVXML***) Batch Control will be duplicated for the Notification Communication Preference MO.

Properties

Property	Property Value
Batch Control	C1USAGER
Description	Resolve XML for C1-USAGE
Template	XML Resolution Template
Selection Algorithm	XML Resolution Select Records
Processing Algorithm	XML Resolution Process Record

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Detailed Description	This batch process resolves foreign keys that may reside in an XML storage field on any table of the specified maintenance object. An XML storage field is either defined as CLOB or XML data type. Refer to the Process Record algorithm for more information about resolution rules. Errors are logged onto the conversion error table. Note that at the start of this job, all rows in the conversion error table for the process maintenance object are deleted. This process is multi-threaded. Records between commits: N/A Minutes between cursor re-initiation: N/A

Parameters

Sequence	Parameter Name	Description	Required	Parameter Value
110	mo	Maintenance Object	Y	C1-USAGE
900	emailMode	Email Mode	N	
910	DIST-THD-POOL	Thread Pool Name	N	
920	maxErrors	Override maximum errors	N	

Generate Keys for C1_USAGE (C1USE01K) (New)

- After creating Batch Controls using the Maintenance Object Configuration, the **Assign Production Keys Template** (***F1-CVASG***) Batch Control will be duplicated for the Notification Communication Preference MO.

Properties

Property	Property Value
Batch Control	C1USE01K
Description	Generate Keys for C1_USAGE
Template	Assign Production Keys Template
Detailed Description	This batch process assigns new keys for the specified table. This process is not multi-threaded. Records between commits: N/A Minutes between cursor re-initiation: N/A

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Parameters

Sequence	Parameter Name	Description	Required	Parameter Value
110	tableId	Table Name	Y	C1_USAGE
120	runType	Run Type	N	
130	startRowNumber	Start from row number	N	
140	parentFieldIds	Parent Field IDs	N	
150	parentFieldDefault	Parent Field Default	N	
160	overrideSqlTimeout	Override SQL Timeout	N	10800
900	emailMode	Email Mode	N	
910	DIST-THD-POOL	Thread Pool Name	N	
920	maxErrors	Override maximum errors	N	

Insert C1_USAGE (C1USE01I) (New)

- After creating Batch Controls using the Maintenance Object Configuration, the **Insert to Production Template** (***F1-CVINS***) Batch Control will be duplicated for the Notification Communication Preference MO.

Properties

Property	Property Value
Batch Control	C1USE01I
Description	Insert C1_USAGE
Template	Insert to Production Template
Detailed Description	This batch process inserts converted data to production for the specified table. If the table is associated with a key table then the batch process also copies the key table. Note that because processing leverages set-based SQL the "Override number of records to commit" must always be 1. Additionally, if "trace output" is enabled, more informative messages get logged onto the batch run tree. These messages help assess the process as the batch progresses. This process is multi-threaded. Records between commits: N/A Minutes between cursor re-initiation: N/A

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Parameters

Sequence	Parameter Name	Description	Required	Parameter Value
110	tableId	Table Name	Y	C1_USAGE
120	commitCount	Number of Commits	N	
130	overrideSqlTimeout	Override SQL Timeout	N	10800
900	emailMode	Email Mode	N	
910	DIST-THD-POOL	Thread Pool Name	N	
920	maxErrors	Override maximum errors	N	

Insert C1_USAGE_CHAR (C1USE02I) (New)

- After creating Batch Controls using the Maintenance Object Configuration, the **Insert to Production Template** (***F1-CVINS***) Batch Control will be duplicated for the Notification Communication Preference MO.

Properties

Property	Property Value
Batch Control	C1USE02I
Description	Insert C1_USAGE_CHAR
Template	Insert to Production Template
Detailed Description	This batch process inserts converted data to production for the specified table. If the table is associated with a key table then the batch process also copies the key table. Note that because processing leverages set-based SQL the "Override number of records to commit" must always be 1. Additionally, if "trace output" is enabled, more informative messages get logged onto the batch run tree. These messages help assess the process as the batch progresses. This process is multi-threaded. Records between commits: N/A Minutes between cursor re-initiation: N/A

Parameters

Sequence	Parameter Name	Description	Required	Parameter Value
110	tableId	Table Name	Y	C1_USAGE_CHAR

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120	commitCount	Number of Commits	N	
130	overrideSqlTimeout	Override SQL Timeout	N	10800
900	emailMode	Email Mode	N	
910	DIST-THD-POOL	Thread Pool Name	N	
920	maxErrors	Override maximum errors	N	

Insert C1_USAGE_LOG (C1USE03I) (New)

- After creating Batch Controls using the Maintenance Object Configuration, the **Insert to Production Template** (***F1-CVINS***) Batch Control will be duplicated for the Notification Communication Preference MO.

Properties

Property	Property Value
Batch Control	C1USE03I
Description	Insert C1_USAGE_LOG
Template	Insert to Production Template
Detailed Description	This batch process inserts converted data to production for the specified table. If the table is associated with a key table then the batch process also copies the key table. Note that because processing leverages set-based SQL the "Override number of records to commit" must always be 1. Additionally, if "trace output" is enabled, more informative messages get logged onto the batch run tree. These messages help assess the process as the batch progresses. This process is multi-threaded. Records between commits: N/A Minutes between cursor re-initiation: N/A

Parameters

Sequence	Parameter Name	Description	Required	Parameter Value
110	tableId	Table Name	Y	C1_USAGE_LOG
120	commitCount	Number of Commits	N	
130	overrideSqlTimeout	Override SQL Timeout	N	10800
900	emailMode	Email Mode	N	
910	DIST-THD-POOL	Thread Pool Name	N	

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920	maxErrors	Override maximum errors	N	

Insert C1_USAGE_LOG_PARM (C1USE04I) (New)

- After creating Batch Controls using the Maintenance Object Configuration, the **Insert to Production Template** (***F1-CVINS***) Batch Control will be duplicated for the Notification Communication Preference MO.

Properties

Property	Property Value
Batch Control	C1USE04I
Description	Insert C1_USAGE_LOG_PARM
Template	Insert to Production Template
Detailed Description	This batch process inserts converted data to production for the specified table. If the table is associated with a key table then the batch process also copies the key table. Note that because processing leverages set-based SQL the "Override number of records to commit" must always be 1. Additionally, if "trace output" is enabled, more informative messages get logged onto the batch run tree. These messages help assess the process as the batch progresses. This process is multi-threaded. Records between commits: N/A Minutes between cursor re-initiation: N/A

Parameters

Sequence	Parameter Name	Description	Required	Parameter Value
110	tableId	Table Name	Y	C1_USAGE_LOG_PARM
120	commitCount	Number of Commits	N	
130	overrideSqlTimeout	Override SQL Timeout	N	10800
900	emailMode	Email Mode	N	
910	DIST-THD-POOL	Thread Pool Name	N	
920	maxErrors	Override maximum errors	N	

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Data Clean-Up / Fixes

- N/A

Recommended Test Cases

Test Case 1: Reported Issue – Conversion process for C1-Usage

- Do the prerequisite steps before running the Conversion Batch Controls
- Run the Conversion Batch Controls
  - Run the Entity Validation – **Validate C1-USAGE** (***C1USAGEV***) Batch Control.
  - Run the XML Resolution - **Resolve XML for C1-USAGE** (***C1USAGER***) Batch Control.
  - Run the Assign Production Keys - **Generate Keys for C1_USAGE** (***C1USE01K***) Batch Control.
  - Run the Added Insertion to Production Batch Controls:
    - **Insert C1_USAGE** (***C1USE01I***).
    - **Insert C1_USAGE_CHAR** (***C1USE02I***).
    - **Insert C1_USAGE_LOG** (***C1USE03I***).
    - **Insert C1_USAGE_LOG_PARM** (***C1USE04I***).

- **Expected Result:** The Batch Controls are successfully processed.

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